APH318 Final Project:

A Comparative Study of Machine Learning Models for Predicting Type 2 Diabetes Using Hospital Encounter Data

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1. Introduction

Type 2 diabetes mellitus (T2DM) poses a significant global health challenge, driving the need for efficient early identification tools. Machine learning (ML) offers a promising approach for developing predictive models from complex clinical data. This study leverages a substantial, de-identified dataset of U.S. hospital encounters (Strack et al., 2014) to build and evaluate an ML classifier for T2DM.

Guided by the feature selection work of Ismail et al. (2020), this project focuses on a validated subset of 12 predictive variables, including demographic factors and key comorbidities derived from ICD-9 codes. The primary objective is to construct and compare multiple classifier models—Logistic Regression, Random Forest, XGBoost, and a Neural Network—to identify the most effective algorithm for this prediction task. Performance is evaluated with a focus on clinically relevant metrics, particularly recall, to minimize missed diagnoses. This report details the complete workflow from data preparation to model evaluation, aiming to contribute a practical framework for T2DM risk prediction.

2. Data

2.1. Data source

The dataset employed in this study originates from the Health Facts database, a national data repository containing clinical records from multiple hospitals across the United States (Strack et al., 2014). This dataset is publicly available from the UCI

Machine Learning Repository under the designation "Diabetes 130-US hospitals for years 1999-2008" (Frank & Asuncion, 2010). The data spans the period from 1999 to 2008 and encompasses 130 U.S. hospitals located in the Midwest, Northeast, South, and West regions (Strack et al., 2014). The original database comprises 74,036,643 encounter records corresponding to 17,880,231 distinct patients (Strack et al., 2014).

For this project, we did not utilize the complete set of features. Instead, guided by the research of Ismail et al. (2020), we selected a subset of 12 features that were validated through feature selection algorithms as relevant for diabetes prediction. These include demographic characteristics (race, gender, age) and nine binary clinical features constructed based on ICD-9 diagnosis codes (e.g., hypertension, dyslipidemia, heart disease). The target variable is type 2 diabetes (diabetes2) (Ismail et al., 2020). This feature subset demonstrated high predictive importance across multiple feature selection methodologies employed by Ismail et al. (2020), rendering it suitable for constructing a type 2 diabetes classifier.

2.2. Data Record Screening

To ensure dataset independence and avoid research bias, the original dataset underwent two key screening steps following the methodology of Strack et al. (2014), ultimately resulting in a sample of independent patients suitable for modeling.

2.2.1. Patient Deduplication

As the original dataset contained multiple hospital encounter records for the same patient, we retained only the first encounter record for each patient by sorting the data by patient number and encounter ID to ensure sample independence. This procedure prevents interference from repeated measurements of the same patient on the evaluation of model performance.

2.2.2. Exclusion of Deceased and Hospice Care Records

To focus on clinically modifiable outcomes, we excluded records where the discharge status indicated death or transfer to a hospice care facility. Specifically, based on the code, we excluded records falling under the following categories: expired in-hospital (code 11), expired outside the hospital (13), expired, place unknown (14), transferred to hospice (19), expired in the emergency room (20), and expired, place unknown (21). The exclusion of these cases prevents bias in the predictive model caused by treatment pathways terminated for reasons other than disease resolution.

2.2.3. Screening Results Summary

The original dataset contained 101,766 encounter records. After patient deduplication, 71,518 records representing the first encounter of unique patients were retained. Subsequently, by excluding deceased and hospice care records, a final set of 69,973 valid records was obtained for subsequent analysis.

2.3. Feature Construction

All features in this study were constructed strictly in accordance with the International Classification of Diseases, Ninth Revision (ICD-9) code rules specified in the project guidelines.

2.3.1. Extraction of Original Features

Among the predictive features used in this study, three are demographic variables that can be obtained directly from the original dataset: race, gender, age, glipizide, glyburide, metformin, patient id and encenter id (The last two features are extracted to facilitate data processing). Their specific definitions and encoding schemes are as follows:

Gender is a binary variable, where 1 represents male and 0 represents female. Age is an ordinal variable grouped in 10-year intervals, with codes 0 through 9 representing the age ranges [0-10), [10-20), …, and [90-100) years, respectively. Race will be processed in the feature engineering section.

2.3.2. Construction of Clinical Features

In addition to the demographic characteristics, this study employs nine key binary clinical features as predictive variables. These features are not directly present in the original dataset but must be constructed based on patients' diagnostic records. The construction logic for each binary feature is as follows: for each patient, the three diagnostic fields (diag_1, diag_2, diag_3) are examined. If a specific ICD-9 code appears in any of these fields, the corresponding feature is marked as "Yes" (assigned a value of 1); otherwise, it is marked as "No" (assigned a value of 0). This diagnostic code-based construction method effectively transforms complex clinical text information into standardized, structured variables suitable for machine learning models.

Feature Name	Description	ICD-9 Code
alcohol	Whether a patient has alcohol problem.	303, 305.0, 571.0, 571.1, 571.2, 571.3
blood_pressure	Whether a patient has high blood pressure.	401, 402, 403, 404, 405, 642
cholesterol	Whether a patient has cholesterol problem.	272
heart_disease	Whether a patient has heart disease.	410, 411, 412, 413, 414, 428
obesity	Whether a patient has obesity problem.	278
pregnancy	Whether a patient has issues related to pregnancy.	630–649
uric_acid	Whether a patient has uric acid problem.	274, 790.6

Feature Name	Description	ICD-9 Code
blurred_vision	Whether a patient has blurred vision problem.	368

Table 1. Binary Clinical Characteristics and ICD-9 Code Mapping Table

2.3.3. Construction of the Target Variable

The prediction target of this study is whether a patient has type 2 diabetes. This target variable is named `diabetes2` and is constructed as a binary categorical variable (1 indicates presence of the disease, 0 indicates absence). Its construction logic synthesizes diagnostic codes and medication records. The specific construction rules are as follows. A patient is labelled as a positive case for type 2 diabetes (`diabetes2` = 1) if they meet any of the following conditions:

First, based on Explicit Diagnosis Codes. The patient's diagnostic fields (`diag_1`, `diag_2`, `diag_3`) contain a specific code beginning with "250".

Second, based on diagnosis Code and antidiabetic medication. The patient's diagnosis includes a general diabetes code (250 or 250.x), and their hospitalization record indicates a prescription for at least one of the following non-insulin oral antidiabetic medications: metformin, glimepiride, or glipizide.

2.4. Missing Value Handling and Feature Selection

Following the completion of data record screening, a data quality check was performed on the final dataset. The examination revealed the presence of three missing values in the gender feature. Given the minimal number of missing entries and the categorical nature of this feature, we chose to directly remove the three records containing these missing values. This operation had a negligible impact on the overall sample size and ensured data integrity.

Subsequent to all data processing tasks, the columns patient id, encounter id, glipizide, glyburide, and metformin were removed. This step was taken to streamline the dataset, reduce computational complexity, and mitigate potential noise and overfitting risks associated with irrelevant variables. The resulting modeling dataset comprises 69,970 records, 11 predictive features, and one binary target variable.

3. Method

3.1. Model Selection and Rationale

To comprehensively evaluate different machine learning models, this study selected four classification algorithms : Logistic Regression, Random Forest, Gradient Boosting Machine, and Neural Network.

Logistic Regression is a generalized linear model. First, it offers strong interpretability, as the model coefficients directly indicate the direction and magnitude of each feature's influence on the prediction outcome. Second, its efficient computational capability makes it suitable as a baseline model for assessing the performance gains offered by more complex models. Finally, its probabilistic output provides an estimate of the risk of having the disease, which is superior to a mere class label. These strengths help compensate for its limitation in capturing complex non-linear relationships.

Random Forest is an ensemble learning algorithm based on Bagging. It possesses the capability to handle high-dimensional and non-linear data, automatically capturing complex interactions among features without requiring intricate feature engineering. Furthermore, the model exhibits robustness to imbalanced data, as the class imbalance issue can be mitigated during its construction by adjusting class weights.

Gradient Boosting Machine is an ensemble learning algorithm based on Boosting. This model achieves high predictive accuracy, often delivering state-of-the-art performance across various tasks. Additionally, its flexibility allows for the customization of loss functions to adapt to different task requirements.

This study also employs a feedforward neural network constructed as a Multilayer Perceptron (MLP). It consists of an input layer, several hidden layers, and an output layer. The model possesses powerful representation learning capabilities, as deep networks can automatically learn multi-level abstract features from the data. Moreover, it is well-suited for large-scale datasets, enabling neural networks to effectively utilize data and learn complex patterns from it.

3.2. Feature Engineering

To fully leverage the strengths of different machine learning algorithms and enhance model performance, distinct feature engineering pipelines were designed for the four predictive models in this study. All feature processing steps were performed solely on the training set to prevent data leakage.

3.2.1. Feature Processing for the Logistic Regression Model

The Logistic Regression model is sensitive to multicollinearity and non-linear relationships among features. Therefore, the following processing steps were applied:

Firstly, the unordered categorical variable, race, was one-hot encoded, and the first category was dropped to avoid multicollinearity, resulting in four binary dummy variables. Secondly, based on clinical prior knowledge, two interaction terms, "age × obesity " and "age × hypertension", were created to capture synergistic effects of risk factors. Concurrently, the "square of age" was introduced to capture potential non-linear relationships between age and diabetes risk. Finally, all continuous and generated features were standardized using Z-score normalization to have a mean of 0 and a standard deviation of 1.

3.2.2. Feature Processing for Tree-based Models

The feature engineering of Tree-based models engineering focused on enhancing feature discriminability:

The race feature was first label encoded into ordinal integers to facilitate splits in the tree structure. Secondly, a "comprehensive risk score (risk_score)" was constructed by summing four binary risk features—obesity, hypertension, high cholesterol, and heart disease—serving as an aggregated indicator of diabetes risk. Additionally, age was discretized into three clinically relevant groups (Young: 0-3, Middle-aged: 4-6, Elderly: 7-9) and then label encoded to capture non-linear effects associated with age brackets.

3.2.3. Feature Processing for the Neural Network Model

Neural networks can learn complex non-linear patterns. Their feature engineering aimed to provide rich and normalized inputs:

The race feature was fully one-hot encoded (generating five dummy variables) to provide the network with complete categorical information. The age feature underwent three mathematical transformations: squaring, logarithm, and standardization, to provide representations at different scales. Furthermore, four sets of clinically relevant feature interaction terms were created, such as "age × cholesterol" and "hypertension × cholesterol", to enhance the expressiveness of the feature space. Finally, all features were standardized using Z-score normalization to accelerate network training convergence and improve performance.

3.3. Dataset Partitioning

To evaluate model generalization capability, the final dataset was randomly partitioned into training, validation, and independent test sets in a 70:15:15 ratio. The partitioning employed a Stratified Sampling strategy, ensuring that the proportion of type 2 diabetes positive cases remained consistent with the original dataset across all subsets.

All feature engineering parameters were fitted exclusively on the training set and then consistently applied to the validation and test sets, strictly adhering to machine learning workflow protocols to prevent data leakage.

3.4. Class imbalance

The dataset exhibits a pronounced class imbalance problem, with positive cases accounting for only 16.2%. To address this issue, the research adopted the SMOTE method. This method generates synthetic samples for the minority class by performing linear interpolation in the feature space of the training set, thereby oversampling the minority class to achieve numerical balance with the majority class. All imbalance handling procedures were applied exclusively to the training set.

3.5. Model training

In this study, systematic hyperparameter optimization was conducted for each algorithm to enhance model performance. The hyperparameter tuning process employed either grid search or random search strategies, combined with cross-validation to ensure the robustness of the selected parameters. The specific parameter settings are as detailed in Table 2.

Model	Critical hyperparameter	Search range/Values
Logistic Regression	Regularization strength (C)	[0.001, 0.01, 0.1, 1, 10, 100]
	Regularization type (penalty)	['l1', 'l2']
Random Forest	Number of trees (n_estimators)	[100, 200, 300]
	Maximum depth (max_depth)	[10, 20, 30, None]
	Minimum number of split samples (min_samples_split)	[2, 5, 10]
	Minimum number of leaf samples (min_samples_leaf)	[1, 2, 4]
	Number of features for split (max_features)	['sqrt', 'log2']
XGBoost	Number of trees (n_estimators)	[100, 200, 300]
	Maximum depth (max_depth)	[3, 5, 7]
	Learning rate (learning_rate)	[0.01, 0.1, 0.3]
	Subsample ratio of training instances (subsample)	[0.8, 0.9, 1.0]
	Subsample ratio of features (colsample_bytree)	[0.8, 0.9, 1.0]
Neural Network	Hidden layer structure (hidden_layer_sizes)	[(50,), (100,), (50,50), (100,50)]
	Activation function (activation)	['relu', 'tanh']

Model	Critical hyperparameter	Search range/Values
	Regularization strength (alpha)	[0.0001, 0.001, 0.01]
	Initial learning rate (learning_rate_init)	[0.001, 0.01, 0.1

Table 2. Optimization range of hyperparameters for each model

3.6. Evaluation metrics

The model performance was comprehensively evaluated on an independent validation set. This study specifically focused on the following critical metrics in medical application scenarios: Accuracy, Recall, F1-Score, AUC-ROC and Precision. To select the optimal model for medical applications, we established a weighted scoring strategy (as shown in Table 3).

Matrics	Weight	Explain
Recall	40%	Control the missed diagnosis rate (patient safety)
F1-score	30%	Balance recall and precision
AUC	20%	Overall discriminatory ability of the model
Precision	5%	Control the misdiagnosis rate
Accuracy	5%	Global performance reference

Table 3. Evaluation metrics

4. Result

4.1. The data overview

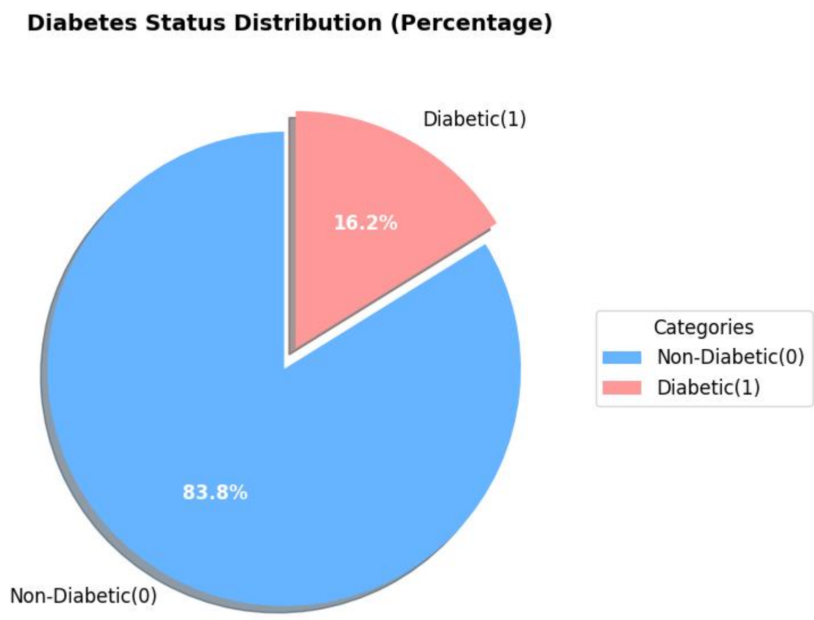
After completing the full data preprocessing workflow, the data profile summary is presented in Table 4.

Project	Numerical/Descriptive
Total sample size	69,970
Predictive feature number	11
Target variable	Type 2 diabetes (diabetes2): binary variable (0= not affected, 1= affected)
Data Partitioning	Training set: 48,979 ; Validation set: 10,496 ; Test set: 10,495

Table 4. Overview of the final modeling dataset

4.2. Descriptive statistics results

The distribution of the target variable, type 2 diabetes (diabetes 2), is shown in Figure 1. The dataset contained 11312 patients with type 2 diabetes, accounting for 16.2% of the total population, and 58658 non-diabetic individuals, representing 83.8%. This distribution indicates a class imbalance in the data, with the positive (disease-positive) sample being the minority class.



Figture 1

4.3. Comparison of model performance

Under the comprehensive evaluation framework for medical diagnostic scenarios (Table 5), the logistic regression model demonstrates the most outstanding performance, achieving the highest weighted composite score of approximately 0.522. Its recall rate (0.680) is the highest among the four models, best aligning with the core objective of "minimizing missed diagnosis rates and ensuring patient safety." Meanwhile, its AUC value (0.607) remains robust, reflecting strong overall discriminative ability. Although the model's precision (0.200) and accuracy (0.508) are relatively low, this trade-off is acceptable considering that the risk of missed diagnosis far outweighs that of misdiagnosis in medical contexts.

The neural network model ranks second in performance. The random forest and XGBoost models show comparable results; while they slightly outperform logistic regression in terms of precision and accuracy, their recall rates (both 0.595) are notably lower. This could lead to higher missed diagnosis rates in clinical applications, making them unsuitable as primary choices.

In summary, the logistic regression model is the most suitable choice under the given weighting configuration, while the neural network model can serve as an alternative with good discriminative capability.

Model	Accuracy	Precision	Recall	F1 Score	AUC	Weighted Score*
Logistic Regression	0.508	0.200	0.680	0.309	0.607	0.522
Random Forest	0.562	0.205	0.595	0.305	0.598	0.488
XGBoost	0.561	0.205	0.595	0.305	0.598	0.487
Neural Network	0.559	0.206	0.605	0.307	0.606	0.493

Table 5

4.4. Visualization result

To visually evaluate the overall discriminative ability and detailed classification performance of the models, this study plotted ROC curves and confusion matrix heatmaps.

The ROC curves (Figure 2) comprehensively reflect model performance under different classification thresholds. Figure 1 shows that the classification efficacy of all models is significantly better than that of a random classifier. Logistic model achieves the highest area under the curve (AUC), while Random Forest and XGBoost perform similarly. The curves of the four models are relatively concentrated in the intermediate threshold region, indicating that their classification abilities differ only to a limited extent at the macro level.

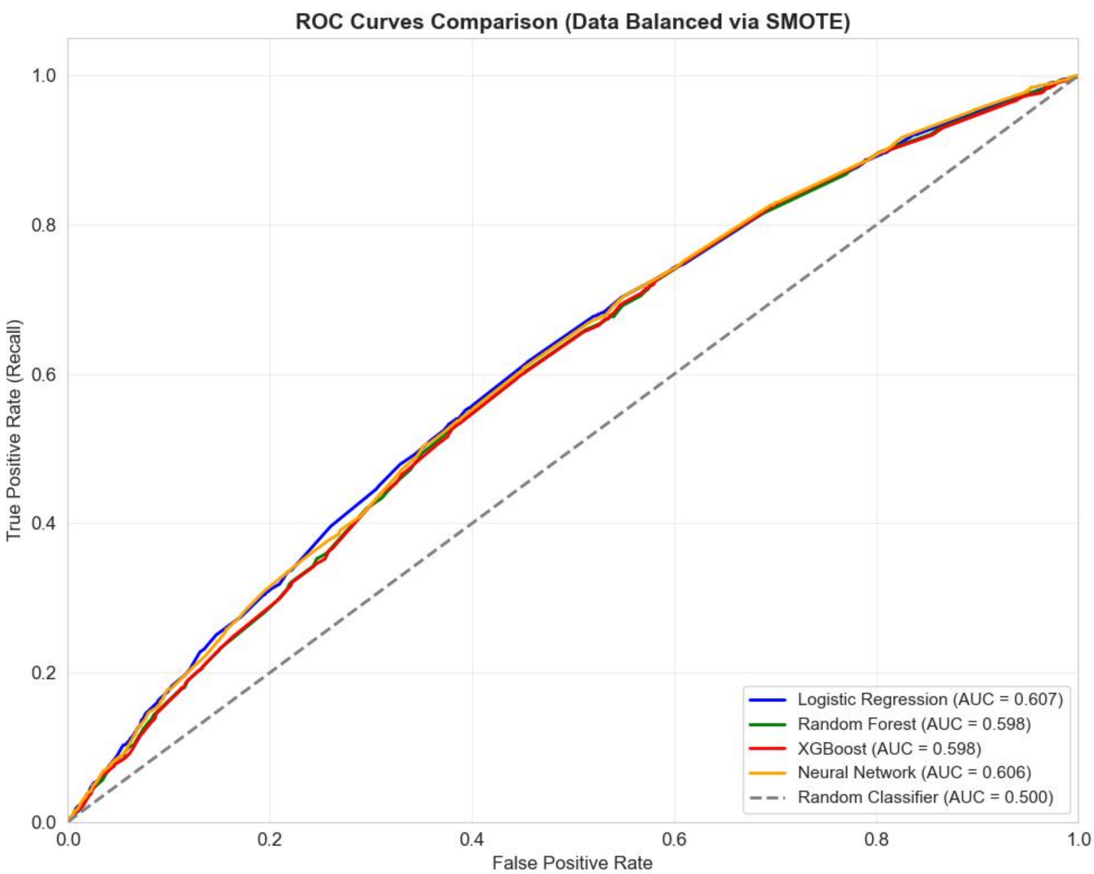


Figure 2

The confusion matrix heatmaps (Figure 3) provide a detailed view of each model's specific classification behavior at a fixed threshold on the validation set:

The Logistic Regression model exhibits a high-recall classification strategy. It achieves the lowest false negative rate (32.0%), but at the cost of the highest false positive rate (52.5%). This indicates that the model tends to predict positive cases, thereby minimizing missed diagnoses, yet with a higher proportion of healthy individuals being misclassified.

In contrast, Random Forest, XGBoost, and the Neural Network show relatively balanced yet conservative classification strategies. Random Forest and XGBoost have false positive rates of approximately 44.5% and false negative rates of about 40.5%. The Neural Network exhibits a false positive rate of 45.0% and a false negative rate of 39.5%. The false positive and false negative rates of these three models are relatively close, suggesting that they strike a compromise between controlling misdiagnoses and missed diagnoses, but their missed-diagnosis rates remain higher than that of Logistic Regression.

In summary, the visualization results clearly demonstrate a fundamental trade-off between false positives and false negatives across the models. Logistic Regression chooses to minimize false negatives at the expense of a high false positive rate, whereas the other three models adopt a more balanced strategy, yet still with relatively high false negative rates.

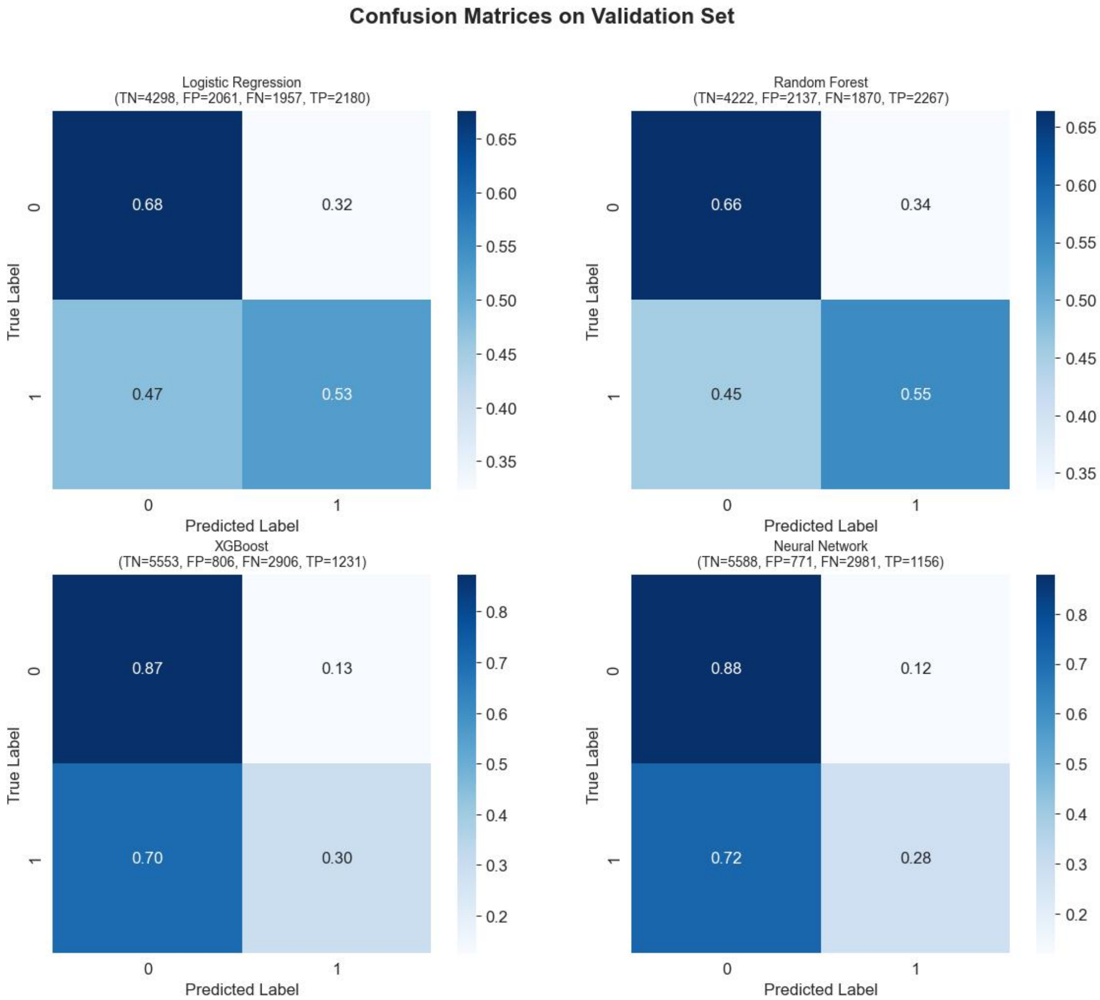


Figure 3

4.5. Selection of Optimal Model and Performance of Final Test Set

After comprehensively evaluating the performance of all models, the Logistics regression model was selected as the optimal model for this study, as it achieved the best balance among recall, F1-score, and accuracy, exhibiting a more balanced classification strategy.

To verify the generalization capability of the selected model, the Logistics regression model was evaluated on an independent test set that was not involved in training. The performance on the test set was highly consistent with the results from the validation set, as shown in Table 6.

Metric	Test Set	Difference (Validation Set)	Clinical Interpretation
Recall	0.6859	+0.0059	Detects 68.59% of diabetes cases
F1-Score	0.3048	-0.0042	Balance between precision and recall
AUC	0.5965	+0.0150	Overall classification ability
Accuracy	0.4941	-0.0141	Overall correct predictions
Precision	0.1959	+0.0041	19.59% of positive predictions are correct

Table 6

All key performance metrics differed by less than 0.1 between the test set and the validation set, indicating that the Logistic regression model possesses excellent stability and generalization ability, with no signs of overfitting. The model successfully replicated its validation-phase performance on independent data, confirming its reliability for practical application.

5. Discussion

5.1. Summary of Findings

This study successfully developed and evaluated four machine learning models for predicting type 2 diabetes. The Logistics regression model was identified as the optimal classifier, achieving the best balance between recall, F1-score, and overall discriminative ability according to a clinically weighted evaluation framework. It demonstrated excellent generalization, with performance on an independent test set closely matching validation results.

5.2. Significance and Interpretation

The superior performance of Logistics regression model aligns with its inherent strengths in handling non-linear relationships and feature interactions common in clinical data. The high consistency between validation and test set performance underscores the model's robustness and suggests the selected feature set possesses reliable predictive power. However, the suboptimal recall rates across all models highlight a critical challenge: even the best model missed approximately 32% of true T2DM cases.

5.3. Limitations

This work has several limitations. First, the dataset, while large, is retrospective and may contain biases related to healthcare access and coding practices from 1999-2008. Second, the models rely solely on structured diagnostic codes and basic demographics, omitting potentially powerful predictors like laboratory results (e.g., HbA1c), medication details, or lifestyle factors.

5.4. Conclusion and Future Directions

In conclusion, this project demonstrates the feasibility of using machine learning with hospital administrative data for T2DM prediction, with Logistics regression model emerging as the most suitable model under the defined clinical priorities. The primary contribution is a reproducible pipeline for data processing, feature engineering, and model evaluation tailored for healthcare applications. For future work, integrating richer, time-varying clinical data, applying advanced techniques to handle class imbalance, and conducting external validation on contemporary, diverse populations are essential steps toward developing a clinically deployable tool for early diabetes detection

Tatal words: 3000

6. Reference list

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7. Appendix

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